

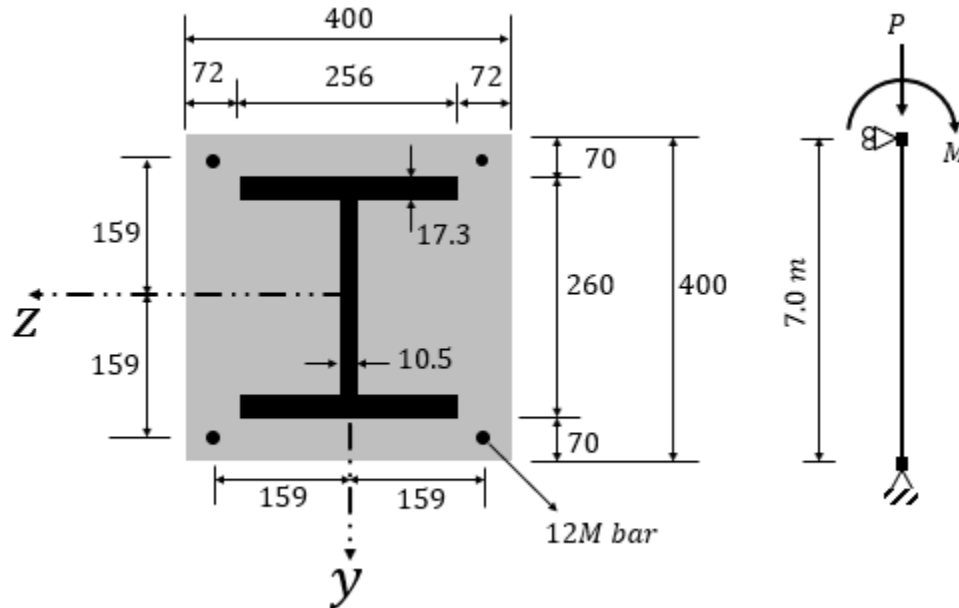
IS 11384:2022 CCD Example 001

CONCRETE-ENCASED RECTANGULAR COMPOSITE COLUMN

EXAMPLE DESCRIPTION

Compression, moment, and shear capacities and demand/capacity ratio are calculated for concrete-encased rectangular section member at mid-height as shown below. It is simply supported with a length of 7.0 meters.

GEOMETRY, PROPERTIES AND LOADING



(The dimensions of the cross-section are in mm.)

Dead: $P = 1200 \text{ kN}$

Live: $P = 600 \text{ kN}, M_y = 380 \text{ kN} - \text{m}, M_z = 50 \text{ kN} - \text{m}$

TECHNICAL FEATURES TESTED

- Axial compressive strength
- Major moment strength
- Shear strength
- Demand/Capacity ratio.

COMPUTER FILE: IS 11384 2022 CCD Ex001

Applicable Programs

➤ ETABS

RESULTS COMPARISON

Independent results are hand calculated.

CONCLUSION

The results show excellent match with independent results.

Benchmarks: ETABS

Output Parameter	Independent	Program	Percent Difference
Axial - P_d (kN)	5701	5701	0.00%
Axial - $P'_{d,C,z}$ (kN)	1837	1837	0.00%
Axial - $P'_{d,C,y}$ (kN)	3859	3859	0.00%
Axial reduction factor - χ_z	0.76	0.76	0.00%
Axial reduction factor - χ_y	0.62	0.62	0.00%
Flexure - $M_{max,z}$ (kN - m)	505	505	0.00%
Flexure - M_{dz} (kN - m)	468	468	0.00%
Flexure - $M_{max,y}$ (kN - m)	310	310	0.00%
Flexure - M_{dy} (kN - m)	189	189	0.00%
Shear - V_{dy} (kN)	544	544	0.00%
Shear - V_{dz} (kN)	1764	1764	0.00%
D/C - Combined Axial and Biaxial Bending	1.377	1.377	0.00%

HAND CALCULATION

Properties:

Material:

Steel: Fe345

$$E_s = 210,000 \text{ N/mm}^2, f_y = 345 \text{ N/mm}^2, \gamma_{m0} = 1.1$$

Concrete: M30

$$E_{cm} = 27,386 \text{ N/mm}^2, f_{ck} = 30 \text{ N/mm}^2$$

$$\gamma_c = 1.5, \alpha_c = 0.85, \alpha_{cc} = 1.0, \eta = 1$$

Reinforcement: HYSD415

$$E_{st} = 200,000 \text{ N/mm}^2, f_{yk} = 415 \text{ N/mm}^2, \gamma_k = 1.15$$

Member: $K_y = K_z = 1.0$ for a pinned-pinned condition
 $L_y = L_z = 7000 \text{ mm}$

Loadings: Dead: $P = 1200 \text{ kN}$
 Live: $P = 600 \text{ kN}, M_y = 380 \text{ kN} - m, M_z = 50 \text{ kN} - m$

Required strengths: for the section in the middle

$$P = P_D + P_L = 1200 + 600 = 1800 \text{ (kN)}$$

$$M_z = \frac{380}{2} = 190 \text{ (kN)}$$

$$M_y = \frac{50}{2} = 25 \text{ (kN)}$$

$$V_y = \frac{M_z}{L} = \frac{380}{7.0} = 54.3 \text{ (kN)}$$

$$V_z = \frac{M_y}{L} = \frac{50}{7.0} = 7.1 \text{ (kN)}$$

Section properties:

Embedded steel I-section:

$$A_s = 112.2 \times 10^2 \text{ (mm}^2\text{)}$$

$$I_{s,z} = 14068 \times 10^4 \text{ (mm}^4\text{)}$$

$$I_{s,y} = 4840 \times 10^4 \text{ (mm}^4\text{)}$$

$$Z_{ps,z} = 1208 \times 10^3 \text{ (mm}^3\text{)}$$

$$Z_{ps,y} = 573 \times 10^3 \text{ (mm}^3\text{)}$$

Reinforcement:

$$A_{st} = 4 \times 113 = 452 \text{ mm}^2$$

$$I_{sti} = \frac{\pi d_b^4}{64} = \frac{\pi (12)^4}{64} = 1017.9 \text{ mm}^4$$

$$I_{st,z} = I_{st,y} = \sum_{i=1}^4 I_{sri} + \sum_{i=1}^4 A_{sti} e_i^2 = 4 \times 1017.9 + 4 \times 113 \times 159^2$$

$$= 1144 \times 10^4 \text{ (mm}^4\text{)}$$

$$Z_{pr,z} = Z_{pr,y} = \sum_{i=1}^4 A_{sti} e_i = 4 \times 113 \times 159 = 72 \times 10^3 \text{ (mm}^3\text{)}$$

Concrete:

$$A_c = b_c h_c - A_s - A_{st} = 400 \times 400 - 112.2 \times 10^2 - 452 = 1483.2 \times 10^2 \text{ (mm}^2\text{)}$$

$$I_{c,z} = \frac{b_c h_c^3}{12} - I_{s,z} - I_{st,z} = \frac{400 \times 400^3}{12} - 14068 \times 10^4 - 1144 \times 10^4$$

$$= 198122 \times 10^4 \text{ (mm}^4\text{)}$$

$$I_{c,y} = \frac{h_c b_c^3}{12} - I_{s,y} - I_{st,y} = \frac{400 \times 400^3}{12} - 4840 \times 10^4 - 1144 \times 10^4$$

$$= 207350 \times 10^4 \text{ (mm}^4\text{)}$$

$$Z_{pc,z} = \frac{b_c h_c^2}{4} - Z_{pa,z} - Z_{pr,z} = \frac{400 \times 400^2}{4} - 1208 \times 10^3 - 72 \times 10^3$$

$$= 14720 \times 10^3 \text{ (mm}^3\text{)}$$

$$Z_{pc,y} = \frac{h_c b_c^2}{4} - Z_{pa,y} - Z_{pr,y} = \frac{400 \times 400^2}{4} - 573 \times 10^3 - 72 \times 10^3$$

$$= 15355 \times 10^3 \text{ (mm}^3\text{)}$$

Simplified calculation method for the interaction curve:

$$P_{dd} = P_D = 1200 \text{ kN}$$

$$P = P_D + P_L = 1200 + 600 = 1800 \text{ kN}$$

$$(EI)_{e,z} = E_s I_{s,z} + 0.6 E_{cm} I_{c,z} + E_{st} I_{st,z}$$

$$= 210,000 \times 14068 \times 10^4 + 0.6 \times 27,386 \times 198122 \times 10^4$$

$$+ 200,000 \times 1144 \times 10^4 = 64385 \text{ kN} - \text{m}^2$$

$$(EI)_{e,y} = E_s I_{s,y} + 0.6 E_{cm} I_{c,y} + E_{st} I_{st,y}$$

$$= 210,000 \times 4840 \times 10^4 + 0.6 \times 27,386 \times 207350 \times 10^4$$

$$+ 200,000 \times 1144 \times 10^4 = 46522 \text{ kN} - \text{m}^2$$

$$P_{cr,z} = \frac{\pi^2 (EI)_{e,z}}{(K_z L_z)^2} = \frac{\pi^2 \times 64385}{(7.0)^2} = 12968 \text{ kN}$$

$$P_{cr,y} = \frac{\pi^2 (EI)_{e,y}}{(K_y L_y)^2} = \frac{\pi^2 \times 46522}{(7.0)^2} = 9371 \text{ kN}$$

$$P_n = A_s f_y + A_{st} f_{yk} + 0.8 A_c \alpha_c f_{ck}$$

$$= 112.2 \times 10^2 \times 345 + 4 \times 113 \times 415 + 0.8 \times 1483.2 \times 10^2 \times 0.85 \times 30$$

$$= 7086 \text{ kN}$$

$$\begin{aligned}
 P_d &= A_s \frac{f_y}{\gamma_{m0}} + A_{st} \frac{f_{yk}}{\gamma_k} + 0.68 A_c \frac{f_{ck}}{\gamma_c} \\
 &= 112.2 \times 10^2 \times \frac{345}{1.1} + 452 \times \frac{415}{1.15} + 0.68 \times 1483.2 \times 10^2 \times \frac{30}{1.5} = 5701 \text{ kN} \\
 \delta &= \frac{A_s f_y}{P_d \gamma_{m0}} = \frac{112.2 \times 10^2 \times 345}{5701 \times 10^3 \times 1.1} = 0.62 \rightarrow 0.2 < \delta = 0.62 < 0.9 \\
 \lambda_z &= \sqrt{\frac{P_n}{P_{cr,z}}} = \sqrt{\frac{7086}{12968}} = 0.74 \\
 \lambda_y &= \sqrt{\frac{P_n}{P_{cr,y}}} = \sqrt{\frac{7086}{9371}} = 0.87
 \end{aligned}$$

As $0.2 < \delta = 0.62 < 0.9$ and $\lambda_z = 0.74 < 2.0$ and $\lambda_y = 0.87 < 2.0 \rightarrow$ the simplified method can be used.

Buckling about z-axis:

Assume the neutral axis is in the web, i.e. $h_{n,z} \leq \frac{h}{2} - t_f \rightarrow A_{stc} = 2 \times 113 = 226 \text{ mm}^2$

$$\begin{aligned}
 h_{n,z} &= \frac{0.8 A_c \alpha_c \frac{f_{ck}}{\gamma_c} - A_{stc} \left(\frac{2 f_{yk}}{\gamma_k} - 0.8 \alpha_c \frac{f_{ck}}{\gamma_c} \right)}{1.6 b_c \alpha_c \frac{f_{ck}}{\gamma_c} + 2 t_w \left(\frac{2 f_y}{\gamma_{m0}} - 0.8 \alpha_c \frac{f_{ck}}{\gamma_c} \right)} \\
 &= \frac{0.8 \times 1483.2 \times 10^2 \times 0.85 \times \frac{30}{1.5} - 226 \left(\frac{2 \times 415}{1.15} - 0.8 \times 0.85 \times \frac{30}{1.5} \right)}{1.6 \times 400 \times 0.85 \times \frac{30}{1.5} + 2 \times 10.5 \left(\frac{2 \times 345}{1.1} - 0.8 \times 0.85 \times \frac{30}{1.5} \right)} = 78.6 \text{ mm}
 \end{aligned}$$

$\rightarrow h_{n,z} = 78.6 < 112.7 = \frac{260}{2} - 17.3 = \frac{h}{2} - t_f \rightarrow$ the assumption on the neutral axis is valid

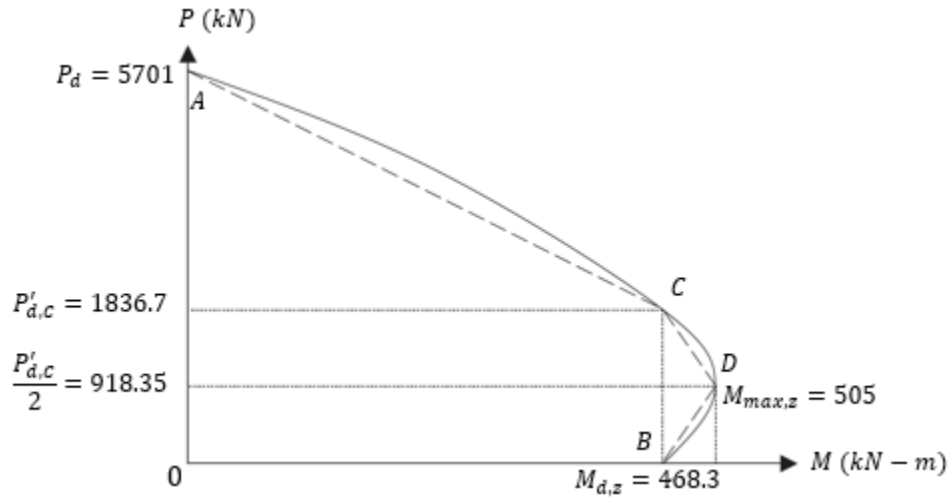
$$Z_{prn,z} = \sum_{i=1}^0 A_{sti} e_i = 0 \text{ (mm}^3\text{)}$$

$$Z_{psn,z} = t_w h_{n,z}^2 = 10.5 \times 78.6^2 = 64.8 \times 10^3 \text{ mm}^3$$

$$Z_{pcn,z} = b_c h_{n,z}^2 - Z_{psn,z} = 400 \times 78.6^2 - 64.8 \times 10^3 = 2404 \times 10^3 \text{ mm}^3$$

$$\begin{aligned}
 P'_{d,c} &= 1.6 h_{n,z} b_c \alpha_c \frac{f_{ck}}{\gamma_c} + 4 h_{n,z} t_w \left(\frac{f_y}{\gamma_{m0}} - 0.8 \frac{f_{ck}}{\gamma_c} \right) \\
 &= 1.6 \times 78.6 \times 400 \times 0.85 \times \frac{30}{1.5} + 4 \times 78.6 \times 10.5 \left(\frac{345}{1.1} - 0.8 \frac{30}{1.5} \right) \\
 &= 1836.7 \text{ kN}
 \end{aligned}$$

$$\begin{aligned}
 M_{d,z} &= (Z_{ps,z} - Z_{psn,z}) \frac{f_y}{\gamma_{m0}} + (Z_{pr,z} - Z_{prn,z}) \frac{f_{yk}}{\gamma_k} + 0.4\alpha_c (Z_{pc,z} - Z_{pcn,z}) \frac{f_{ck}}{\gamma_c} \\
 &= (1208 \times 10^3 - 64.8 \times 10^3) \frac{345}{1.1} + (72 \times 10^3 - 0) \frac{415}{1.15} \\
 &\quad + 0.4 \times 0.85 (14720 \times 10^3 - 2404 \times 10^3) \frac{30}{1.5} = 468.3 \text{ kN-m} \\
 M_{max,z} &= Z_{ps,z} \frac{f_y}{\gamma_{m0}} + Z_{pr,z} \frac{f_{yk}}{\gamma_k} + 0.4\alpha_c Z_{pc,z} \frac{f_{ck}}{\gamma_c} \\
 &= 1208 \times 10^3 \frac{345}{1.1} + 72 \times 10^3 \frac{415}{1.15} + 0.4 \times 0.85 \times 14720 \times 10^3 \frac{30}{1.5} \\
 &= 505 \text{ kN-m}
 \end{aligned}$$



Buckling about y-axis:

Assume the neutral axis is in the web, i.e. $\frac{t_w}{2} < h_{n,y} < \frac{b_f}{2} \rightarrow A_{stc} = 2 \times 113 = 226 \text{ mm}^2$

$$\begin{aligned}
 h_{n,y} &= \frac{0.8A_c \frac{f_{ck}}{\gamma_c} - A_{stc} \left(\frac{2f_{yk}}{\gamma_k} - 0.8 \frac{f_{ck}}{\gamma_c} \right) - t_w (2t_f - h) \left(\frac{2f_y}{\gamma_{m0}} - 0.8 \frac{f_{ck}}{\gamma_c} \right)}{1.6h_c \frac{f_{ck}}{\gamma_c} + 4t_w \left(\frac{2f_y}{\gamma_{m0}} - 0.8 \frac{f_{ck}}{\gamma_c} \right)} \\
 &= \frac{0.8 \times 1483.2 \times 10^2 \times \frac{30}{1.5} - 226 \left(\frac{2 \times 415}{1.15} - 0.8 \frac{30}{1.5} \right) - 10.5 (2 \times 17.3 - 260) \left(\frac{2 \times 345}{1.1} - 0.8 \frac{30}{1.5} \right)}{1.6 \times 400 \times \frac{30}{1.5} + 4 \times 10.5 \left(\frac{2 \times 345}{1.1} - 0.8 \times \frac{30}{1.5} \right)} \\
 &= 94.4 \text{ mm}
 \end{aligned}$$

$\rightarrow \frac{t_w}{2} = \frac{10.5}{2} = 5.25 < h_{n,y} = 94.4 < 128 = \frac{256}{2} = \frac{b_f}{2} \rightarrow$ the assumption on the neutral axis is valid

$$Z_{prn,y} = \sum_{i=1}^0 A_{sti} e_i = 0 \text{ (mm}^3\text{)}$$

$$Z_{psn,y} = 2t_f h_{n,y}^2 + \frac{(h - 2t_f)t_w^2}{4} = 2 \times 17.3 \times 94.4^2 + \frac{(260 - 2 \times 17.3)10.5^2}{4}$$

$$= 314.8 \times 10^3 \text{ mm}^3$$

$$Z_{pcn,y} = h_c h_{n,y}^2 - Z_{psn,y} = 400 \times 94.4^2 - 314.8 \times 10^3 = 3253 \times 10^3 \text{ mm}^3$$

$$P'_{d,c} = 1.6h_{n,y}h_c \frac{f_{ck}}{\gamma_c} + [4h_{n,y}t_f + t_w(h - 2t_f)] \left(\frac{f_y}{\gamma_{m0}} - 0.8 \frac{f_{ck}}{\gamma_c} \right)$$

$$= 1.6 \times 94.4 \times 400 \frac{30}{1.5}$$

$$+ [4 \times 94.4 \times 17.3 + 10.5(260 - 2 \times 17.3)] \left(\frac{345}{1.1} - 0.8 \frac{30}{1.5} \right)$$

$$= 3858.6 \text{ kN}$$

$$M_{d,y} = (Z_{ps,y} - Z_{psn,y}) \frac{f_y}{\gamma_{m0}} + (Z_{pr,y} - Z_{prn,y}) \frac{f_{yk}}{\gamma_k} + 0.4\alpha_c (Z_{pc,y} - Z_{pcn,y}) \frac{f_{ck}}{\gamma_c}$$

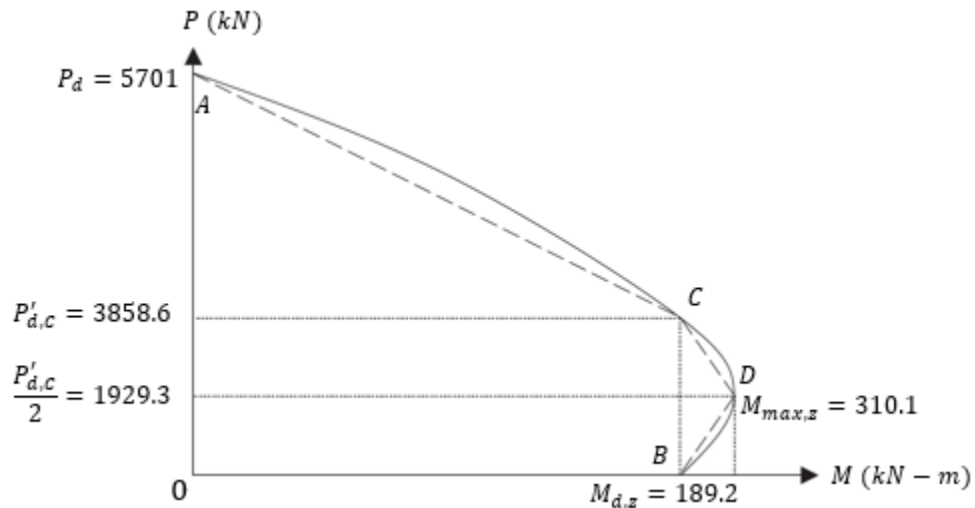
$$= (573 \times 10^3 - 314.8 \times 10^3) \frac{345}{1.1} + (72 \times 10^3 - 0) \frac{415}{1.15}$$

$$+ 0.4 \times 0.85 (15355 \times 10^3 - 3253 \times 10^3) \frac{30}{1.5} = 189.2 \text{ kN-m}$$

$$M_{max,y} = Z_{ps,y} \frac{f_y}{\gamma_{m0}} + Z_{pr,y} \frac{f_{yk}}{\gamma_k} + 0.4\alpha_c Z_{pc,y} \frac{f_{ck}}{\gamma_c}$$

$$= 573 \times 10^3 \frac{345}{1.1} + 72 \times 10^3 \frac{415}{1.15} + 0.4 \times 0.85 \times 15355 \times 10^3 \frac{30}{1.5}$$

$$= 310.1 \text{ kN-m}$$



Global buckling under axial load:

Buckling about z-axis:

Buckling curve $b \rightarrow \alpha = 0.34$

$$\Phi = 0.5[1 + \alpha(\lambda_z - 0.2) + \lambda_z^2] = 0.5[1 + 0.34(0.74 - 0.2) + 0.74^2] = 0.865$$

$$\chi_z = \frac{1}{\Phi + \sqrt{\Phi^2 - \lambda_z^2}} = \frac{1}{0.865 + \sqrt{0.865^2 - 0.74^2}} = 0.76$$

$$\frac{P}{\chi_z P_d} = \frac{1800}{0.76 \times 5701} = 0.415$$

Buckling about y-axis:

Buckling curve $c \rightarrow \alpha = 0.49$

$$\Phi = 0.5[1 + \alpha(\lambda_y - 0.2) + \lambda_y^2] = 0.5[1 + 0.49(0.87 - 0.2) + 0.87^2] = 1.042$$

$$\chi_y = \frac{1}{\Phi + \sqrt{\Phi^2 - \lambda_y^2}} = \frac{1}{1.042 + \sqrt{1.042^2 - 0.87^2}} = 0.62$$

$$\frac{P}{\chi_y P_d} = \frac{1800}{0.62 \times 5701} = 0.509$$

Second-Order moments:

Bending about z-axis:

As $\lambda_z = 0.74 > 0.2 \rightarrow$ second-order moment must be considered for bending about z-axis

For the first-order moment, its diagram is linear with $\psi = \frac{M_1}{M_2} = 0$

$$\rightarrow C_{mz} = 0.6 + 0.4\psi = 0.6 > 0.4$$

$$k_z = \frac{C_{mz}}{1 - P/P_{cr,z}} = \frac{0.6}{1 - 1800/12968} = 0.697 < 1.0 \rightarrow k_z = 1.0$$

$$M_z = k_z M_{max,z} = 1.0 \times 380 = 380 \text{ kN} - m$$

Bending about y-axis:

As $\lambda_y = 0.87 > 0.2 \rightarrow$ second-order moment must be considered for bending about y-axis

For the first-order moment, its diagram is linear with $\psi = \frac{M_1}{M_2} = 0$

$$\rightarrow C_{my} = 0.6 + 0.4\psi = 0.6 > 0.4$$

$$k_y = \frac{C_{my}}{1 - P/P_{cr,y}} = \frac{0.6}{1 - 1800/9371} = 0.743 < 1.0 \rightarrow k_y = 1.0$$

$$M_y = k_y M_{max,y} = 1.0 \times 50 = 50 \text{ kN} - m$$

Shear design:

Shear along y-axis:

$$A_{v,y} = ht_w = 260 \times 10.5 = 2730 \text{ mm}^2$$

$$V_{d,y} = A_{v,y} \frac{f_y}{\sqrt{3}} = 2730 \frac{345}{\sqrt{3}} = 544 \text{ kN}$$

$$\frac{V_y}{V_{d,y}} = \frac{54.3}{544} = 0.1 < 0.6$$

→ the effects of shear along y-axis on the moment and axial strengths are ignored.

Shear along z-axis:

$$A_{v,z} = 2bt_f = 2 \times 256 \times 17.3 = 8858 \text{ mm}^2$$

$$V_{d,z} = A_{v,z} \frac{f_y}{\sqrt{3}} = 8858 \frac{345}{\sqrt{3}} = 1764 \text{ kN}$$

$$\frac{V_z}{V_{d,z}} = \frac{7.1}{1764} = 0.004 < 0.6$$

→ the effects of shear along z-axis on the moment and axial strengths are ignored.

Demand/Capacity (D/C) ratio:

$$P_{c,y} = A_c \frac{\alpha_{cc}}{\gamma_c} \eta \lambda_y f_{ck} = 1483.2 \times 10^2 \frac{1.0}{1.5} 1.0 \times 0.87 \times 30 = 2580 \text{ kN}$$

$$P_{c,z} = A_c \frac{\alpha_{cc}}{\gamma_c} \eta \lambda_z f_{ck} = 1483.2 \times 10^2 \frac{1.0}{1.5} 1.0 \times 0.74 \times 30 = 2193 \text{ kN}$$

$$P_{d,y} = P_{d,z} = P_d = 5701 \text{ kN}$$

$$\chi_{c,y} = \frac{P_{c,y}}{P_{d,y}} = \frac{2580}{5701} = 0.4525$$

$$\chi_{c,z} = \frac{P_{c,z}}{P_{d,z}} = \frac{2193}{5701} = 0.3846$$

$$\chi_{d,y} = \frac{P}{P_{d,y}} = \frac{1800}{5701} = 0.3157$$

$$\chi_{d,z} = \frac{P}{P_{d,z}} = \frac{1803}{5701} = 0.3157$$

$$\chi_{d,y} = 0.3157 < 0.4525 = \chi_{c,y}$$

$$\rightarrow \mu_{dy} = 1 - \frac{(1 - \chi_y) \chi_{d,y}}{(1 - \chi_{c,y}) \chi_y} = 1 - \frac{(1 - 0.62) 0.3157}{(1 - 0.4525) 0.62} = 0.6445$$

$$\chi_{d,z} = 0.3157 < 0.3846 = \chi_{c,z}$$

$$\rightarrow \mu_{dz} = 1 - \frac{(1 - \chi_z) \chi_{d,z}}{(1 - \chi_{c,z}) \chi_z} = 1 - \frac{(1 - 0.76) 0.3157}{(1 - 0.3846) 0.76} = 0.8390$$

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$$\begin{aligned}\frac{M_y}{\mu_{dy}M_{dy}} &= \frac{50}{0.6445 \times 189.2} = 0.410 < 0.9 = \alpha_{my} \\ \frac{M_z}{\mu_{dz}M_{dz}} &= \frac{380}{0.8390 \times 468.3} = 0.967 > 0.9 = \alpha_{mz} \\ \frac{M_y}{\mu_{dy}M_{dy}} + \frac{M_z}{\mu_{dz}M_{dz}} &= \frac{50}{0.6445 \times 189.2} + \frac{380}{0.8390 \times 468.3} = 1.377 > 1.0\end{aligned}$$

The D/C ratio of 1.377 governs the design.